

ALFRED NJUGUNA KANYUTU

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Machine Learning Engineer

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alfredportfolio.vercel.app

PROFESSIONAL SUMMARY.

I am a software Engineering graduate who has a strong interest in the development and deployment of machine learning models. I have skills in areas such as designing, training, and deploying ML models using Python, PyTorch, Scikit Learn, FastAPI and Docker. I also have experience in building full stack ML system from data preprocessing to deployment on cloud platforms.

PROJECTS

Dynamic Pricing Model ([Dynamic-pricing-gamma.vercel.app](https://dynamic-pricing-gamma.vercel.app))

- Developed a neural network regression model using PyTorch to predict ride costs based on 1000+ historical ride records. The model was able to achieve a MSE of 20487.5332 and R^2 of 0.4992 on test data.
- Built and deployed a full-stack application using FastAPI(backend) and HTML/CSS/JavaScript(frontend) and connected them via REST APIs.
- Containerized with Docker and deployed backend to Render and frontend to vercel.

Credit Card Fraud Detection System ([Credit-card-fraud-detection-system-delta.vercel.app](https://credit-card-fraud-detection-system-delta.vercel.app))

- Trained a random forest ensemble model to classify if a transaction is fraudulent or not. The model was able to attain a 99.81% accuracy on test data.
- Created the REST API backend using FastAPI, dockerized it and deployed it on render.
- Vercel was used to host the system frontend which was developed using react.

Energy Consumption Prediction System ([Energy-peach-zeta.vercel.app](https://energy-peach-zeta.vercel.app))

- Trained a regression model on 5000+ records to predict energy consumption in a household.
- Developed the REST API backend using FastAPI, dockerized it and deployed it on render.
- Vercel was used to host the system frontend which was created using react.

EDUCATION

- **Oct 2020 – Dec 2024: Kisii University**
Bachelor of Science in Software Engineering

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL, JavaScript
- **Machine Learning:** Scikit Learn, PyTorch, TensorFlow
- **Frameworks:** FastAPI, Spring Boot, React
- **DevOps & Deployment:** Docker, Git, REST APIs.
- **Data Tools:** Pandas, Numpy, Matplotlib, Seaborn

LANGUAGES: English, Swahili